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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)
REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCFTQ301 - TECHNOLOGY OF MEAT, FISH AND POULTRY PRODUCTS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any 5 questions

Each question carries 2 marks

1. **Name two personal hygiene practices required for meat handlers. [K] / [CO1]**
   Wash hands thoroughly before handling meat.
   Wear clean protective clothing and keep fingernails short.

2. Define modified atmosphere packaging (MAP) and mention any two gases used in MAP. [K] / [CO2]

MAP is a packaging technique in which the natural atmosphere inside a package is replaced with a controlled gas mixture to extend the shelf life of food.

Two gases used are carbon dioxide (CO_2) and nitrogen (N_2).

3. **Illustrate the distribution of proteins in different parts of poultry meat. [A] / [CO3]**

   Breast meat contains a higher proportion of white muscle fibres, while leg meat contains more red muscle fibres and myoglobin. Both breast and leg muscles contain myofibrillar proteins (actin and myosin), sarcoplasmic proteins and connective tissue proteins.

4. **List any four basic ingredients used in sausage preparation. [K] / [CO2]**

   Lean meat and fat, Salt and spices.

5. **Define rigor mortis. [K] / [CO1]**

   Rigor mortis is the stiffening of muscles after slaughter due to ATP depletion and the formation of stable actin–myosin cross-bridges.
   It occurs during post-mortem changes in meat and affects meat tenderness

6. **List two advantages of meat irradiation. [K] / [CO2]**

   Reduces pathogenic microorganisms and improves food safety.

Delays spoilage and extends the shelf life of meat.

7.Differentiate between fresh water fish and marine fish? [An]/[CO4]

Freshwater fish: Live in rivers, lakes and ponds; examples include rohu and catla.

Marine fish: Live in seas and oceans; examples include sardine and mackerel.

[2x5=10]

Part B

Short Essay Questions

Answer any 5 questions

Each question carries 4 marks

7. **Differentiate between egg white and egg yolk with respect to composition, nutritional value and functional properties. [U] / [CO5]**

   Composition and nutritional value: Egg white mainly contains water and proteins such as ovalbumin, whereas egg yolk contains fats, proteins, phospholipids, vitamins and minerals.

Functional properties: Egg white provides foaming and gel-forming properties, whereas egg yolk acts as an emulsifier due to the presence of lecithin and other phospholipids.

9. Write a note on mechanical tenderization of meat. [K] / [CO1]

Mechanical tenderization involves breaking muscle fibres and connective tissues using blades, needles, pounding or cubing equipment.

It improves meat tenderness and eating quality; however, blade or needle tenderization may transfer surface microorganisms into the interior of the meat

10. Discuss the advantages of hot smoking in meat preservation. [U] / [CO2]

Hot smoking combines heat treatment, drying and smoke deposition to reduce microbial load and improve flavour, colour and texture.

It helps extend shelf life by reducing moisture and microbial growth, although proper cooking, cooling and storage are essential for safety.

11. Describe the preparation of fish silage and its uses? [U] / [CO4]

Fish silage is prepared by mincing fish waste or whole fish, adding an acid such as formic acid, and allowing enzymes to break down the fish tissues into a liquid product.

It is used as an ingredient in animal and aquaculture feeds and as a source of proteins, peptides and nutrients.

12. Describe the major factors considered in grading meat. [U] / [CO1]

Carcass characteristics: Age, sex, carcass weight, conformation, muscle development and fat distribution.

Meat quality: Colour, texture, marbling, tenderness and maturity.

13. Describe the physical changes occurring in meat during cooking. [U] / [CO2]

Muscle proteins denature and coagulate, causing changes in colour, firmness and texture; muscle fibres shrink as the temperature increases.

Fat melts, moisture is lost, collagen gradually converts to gelatin during prolonged moist cooking, and the meat decreases in weight and volume.

14. Implement suitable handling, packaging and storage practices to maintain the quality and safety of eggs during storage. [A] [CO5]

Handling and packaging: Handle eggs gently, reject cracked or dirty eggs, maintain clean conditions, and use clean, protective cartons to prevent breakage and contamination.

Storage: Store eggs under refrigeration at a consistent temperature, preferably around 4°C or below, maintain the cold chain, and follow the recommended shelf life and stock rotation practices.

[4x5=20]

Part C

Long Essay Questions

Answer any 2 questions .Each question carries 10 marks

15. Explain the structural and sanitary features required for hygienic slaughter and meat processing in a modern abattoir. [U] / [CO1]

Structural features: The abattoir should have separate areas for animal reception, lairage, slaughter, dressing, carcass inspection and meat processing, with impervious floors, adequate drainage, ventilation, lighting and a reliable water supply.

Sanitary features: Proper cleaning and disinfection, hygienic equipment, personal protective clothing, pest control, safe waste disposal, potable water and separation of clean and contaminated areas are essential to prevent cross-contamination.

16. Explain the complete procedure of post-mortem inspection of carcasses and edible organs. U] / [CO1]

Carcass examination: Inspect the carcass visually for abnormalities, bruises, lesions, swelling and contamination; examine and palpate relevant lymph nodes and make incisions where required to detect disease.

Organ examination and disposition: Examine the liver, lungs, heart, kidneys, spleen and other edible organs for lesions, parasites and abnormalities; correlate findings with ante-mortem observations and pass, partially condemn or condemn the carcass and affected organs as appropriate.

17. Explain the importance of pH changes after slaughter in relation to PSE and DFD meat. [A] / [CO2]

PSE meat (Pale, Soft and Exudative): Rapid post-mortem glycolysis while the carcass is still warm causes a rapid pH decline, resulting in pale colour, soft texture and excessive drip loss.

DFD meat (Dark, Firm and Dry): Depletion of muscle glycogen before slaughter results in insufficient lactic acid production and a high ultimate pH, producing dark, firm meat with a sticky surface and reduced shelf life due to increased microbial growth.

18. Explain the production, properties, advantages and agricultural application of fish manure? [U] / [CO4]

Production and properties: Fish manure is produced from fish waste by decomposition, fermentation or composting, sometimes with other organic materials; it contains nitrogen, phosphorus, potassium and other nutrients, along with organic matter.

Advantages and agricultural applications: It improves soil fertility, supplies nutrients to crops, supports beneficial soil microorganisms and promotes plant growth; it can be applied as compost or processed organic fertilizer after suitable treatment to ensure hygienic safety.

[10x2 = 20]